Steering a swarm that owns a proof

Simulation 1 (days and nights), setup `task003-symmetric`

LLM-free simulator; report of 15 October 2026; runs of 15 October 2026

Mirrors the Simulation 2 report of the same date: same sections, figures and equations.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (what it read already proves its target) silences it, posts the b facts from its pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1 - \rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at the start and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The runs. Posts read by an agent $q \in \{3, 6, 12\}$; posts read by the controller $q_c \in \{3, 6, 12, 24\}$; facts posted per night $b \in \{1, \ldots, 9\}$ (at most 12, the pool size); memory $\rho \in \{1, 0.75, 0.5, 0.25\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (1,740 settings). Unless stated otherwise: $q = 6$, $q_c = 12$, $b = 1$ (the main setting, which posts about as many facts as Simulation 2 at rate 1), gate on. Gate-off results are in Appendix A.

Summary. The 24 agents together own a proof of A0, and with full memory they keep it: no controller changes the swarm for more than a few days. **Forgetting is what makes the swarm steerable**, and in this simulation the effect peaks at intermediate memory: at $\rho = 0.5$ the false controller raises the share voting A2 by 0.53–0.57 and wins 78% of the votes, against 0.31–0.36 at $\rho = 0.25$ and 0.22–0.58 at $\rho = 0.75$ (Figures 1, 2). **The budget matters only when agents remember well**: at $\rho \leq 0.5$ one fact per night achieves almost everything, and it is five to seven times more effective per fact than nine (Figures 4, 5). **Reading more posts protects the proof when agents remember, and helps the controller when they forget** (Figure 3c). Every controller replaces part of the agents' own proofs (Figure 6). The target is most readable at $\rho = 0.5$ (0.42–0.49 bits from one vote); the efficiency η is 0.56–0.91 and is lowest where the two controllers are most unequal (Figures 7–10). Turning the vote gate off adds 43–98 facts per episode and up to 0.16 in susceptibility (Appendix A). Swarm-level states read up to five times more than one vote, with a lower efficiency (Appendix B).

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose vote on day t is answer a and $N_t = 24$ (every agent votes every day; before the first day nobody has voted and we set $x_{a,0} = 0$),

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G_t^{(0)}(\pi^+)$ for the controller targeting A0 and $G_t \equiv G_t^{(2)}(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\mathrm{A0}, \mathrm{A2}\}$ is drawn by a fair coin. The state X of the swarm on day t is one of:

- **one agent's vote** (main text): the vote of a randomly chosen agent, $X \in \{\mathrm{A0}, \mathrm{A1}, \mathrm{A2}\}$;
- **plurality**: the answer with the most votes (a k-way tie gives $1/k$ to each);
- **S1**: the agents' mean belief in A0, m_{A0}, cut at 0.33, 0.50 and 0.75 (4 values);
- **S3**: $(m_{\mathrm{A0}}, m_{\mathrm{A2}})$, each cut into thirds of $[0, 1]$ (9 values, 6 reachable).

Each episode e of arm s (silent 0, target A0 $+$, target A2 $-$) gives a probability vector $y_s^{(e)}$ over the values of X (for one agent's vote, the episode's vote shares; for the others, a mark on one value), and

$$P_s(x) = \frac{1}{E} \sum_{e=1}^{E} y_s^{(e)}(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x) \log_2[p(x)/q(x)]$,

$$I(Z; X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z; X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z; X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004 for one agent's vote and 0.06 for the other states; where C is finite without smoothing, by at most 0.014. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted on the nights before day t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring budgets,

$$\frac{\Delta G}{\Delta b} = \frac{G(b^{(k+1)}) - G(b^{(k)})}{b^{(k+1)} - b^{(k)}}. \tag{8}$$

2 Votes over time

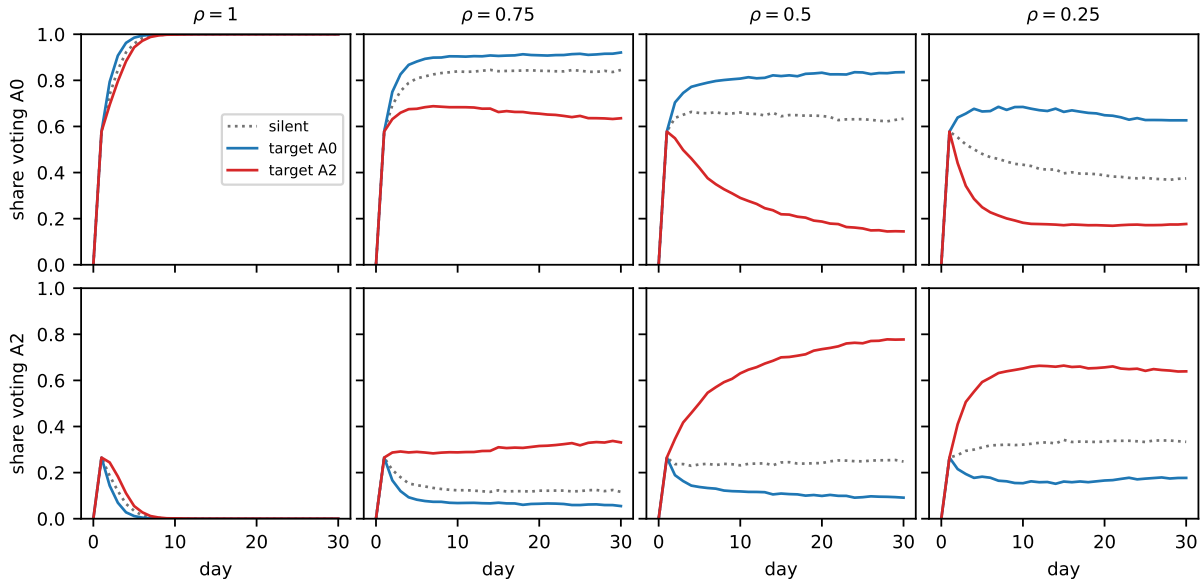


Figure 1: Share of agents voting A0 (top) and A2 (bottom) at the end of each day, Eq. (1), without a controller (dotted), with the controller targeting A0 (blue) and A2 (red); one column per memory ρ. $q = 6$, $q_c = 12$, $b = 1$, gate on. Day 0: nobody has voted yet, every share is 0. 95% intervals are narrower than the lines (± 0.01).

- **The shorter the memory, the less firmly the silent swarm holds the truth**: 100%, 85%, 63% and 38% vote A0 on day 30 for $\rho = 1, 0.75, 0.5, 0.25$.
- **With full memory no controller matters**: every agent votes A0 by day ≈ 8 in all three arms.
- **The false controller wins the majority when** $\rho \leq 0.5$: A2 gets 78% ($\rho = 0.5$) and 64% ($\rho = 0.25$) of the votes, against 25% and 33% in silence; at $\rho = 0.75$ only 33%.
- **The truth controller restores the truth** to 92% ($\rho = 0.75$) and 84% ($\rho = 0.5$), but only to 63% at $\rho = 0.25$.

3 Paired susceptibility over time

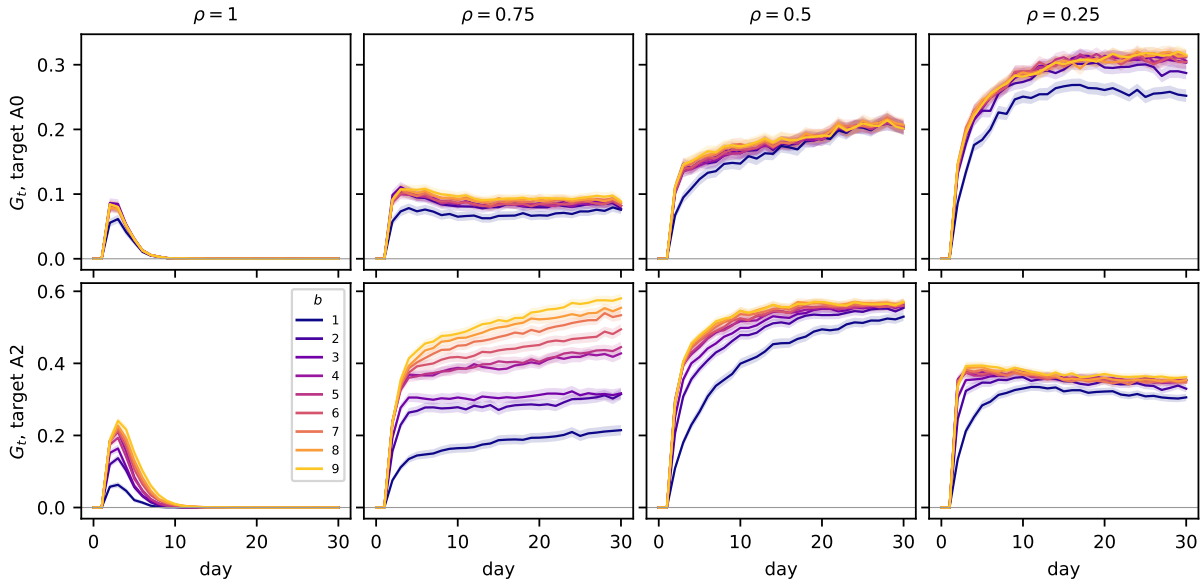


Figure 2: Paired susceptibility G_t, Eq. (2), target A0 (top) and A2 (bottom), one curve per budget b; $q = 6$, $q_c = 12$, gate on. $G_0 = 0$: before the first day the three arms are identical. Bands: 95% intervals.

- **Susceptibility peaks at intermediate memory**: on day 30, target A2, $G = 0.22$–0.58 ($\rho = 0.75$), 0.53–0.57 ($\rho = 0.5$), 0.31–0.36 ($\rho = 0.25$); target A0, 0.08–0.09, 0.20, 0.25–0.32.
- **With full memory control is only transient**: target A2 reaches 0.24 on day 3 ($b = 9$) and is gone by day ≈ 12.
- **The budget matters only when agents remember well**: at $\rho = 0.75$, target A2 grows from 0.22 ($b = 1$) to 0.58 ($b = 9$); at $\rho \leq 0.5$ every $b \geq 2$ gives about the same.
- **With heavy forgetting a large budget overshoots**: at $\rho = 0.25$, $b = 9$, target A2 peaks at 0.39 (day 5) and settles at 0.36.

4 Phase diagrams of the susceptibility

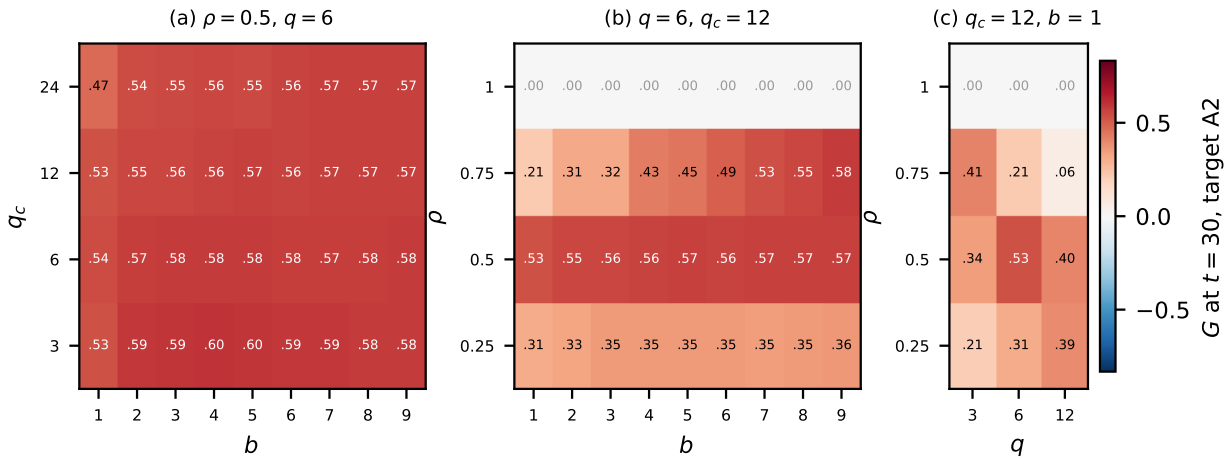


Figure 3: G on day 30, target A2, gate on: (a) over the posts the controller reads q_c and the budget b, at $\rho = 0.5$, $q = 6$; (b) over ρ and b, at $q = 6$, $q_c = 12$; (c) over ρ and the posts each agent reads q, at $q_c = 12$, $b = 1$. Grey number: the 95% interval includes 0. Same colour scale as in the Simulation 2 report.

- **Memory is the main axis** (panel b): $G = 0$ at $\rho = 1$, largest at $\rho = 0.5$ (0.53–0.57), and at $\rho = 0.75$ only with a large budget.
- **At $\rho = 0.5$ the budget and q_c hardly matter** (panel a: 0.47–0.60; mean 0.58, 0.57, 0.56, 0.55 for $q_c = 3, 6, 12, 24$).
- **Reading more defends the proof only when agents remember well** (panel c): at $\rho = 0.75$, G falls from 0.41 ($q = 3$) to 0.06 ($q = 12$); at $\rho = 0.25$ it rises from 0.21 to 0.40. A likely reason, not tested: with heavy forgetting the proof cannot be rebuilt anyway, and reading more only spreads the controller's facts faster.

5 Susceptibility against the budget, and its slope

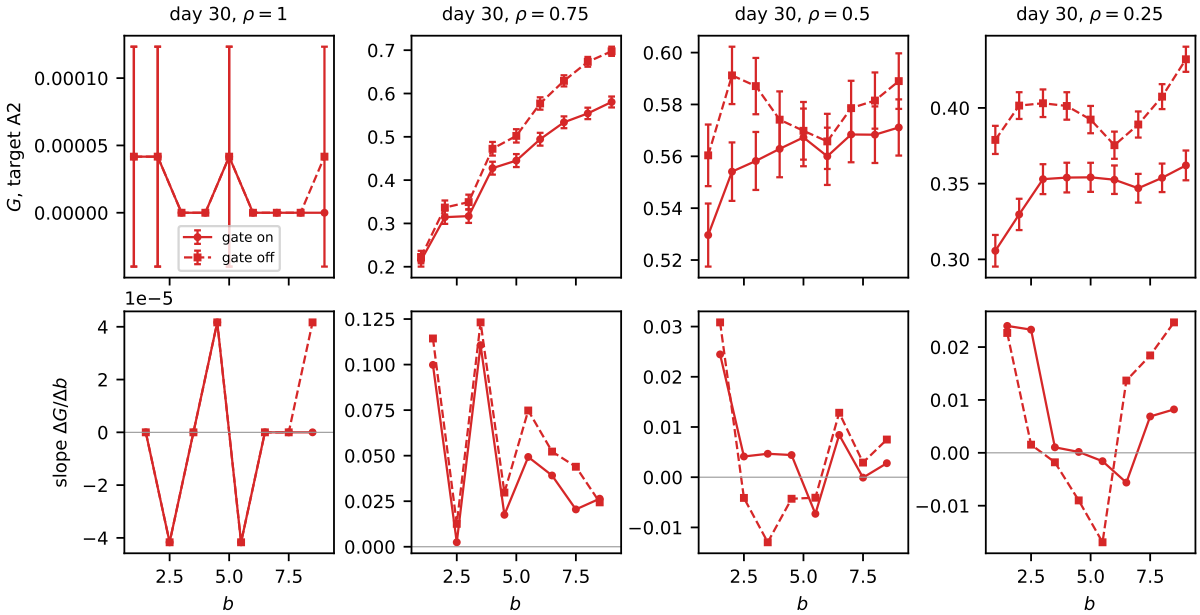


Figure 4: Top: G on day 30, target A2, against the budget b, one panel per ρ; gate on (solid) and off (dashed); bars: 95% intervals. Bottom: the slope between neighbouring budgets, Eq. (8). (Simulation 1 has a single end point, so all four ρ are shown; Simulation 2 shows $t = 5$ and $t = 30$.)

- **When agents remember well, more budget keeps buying susceptibility**: at $\rho = 0.75$ the slope is 0.10 for the first extra fact per night and still 0.04 on average after it.
- **When they forget, one fact per night is almost enough**: at $\rho = 0.5$ and 0.25 the slope is 0.02 for the first step and about 0.002 after it.
- **The marginal susceptibility is the efficiency of adding budget**: worth paying for only at $\rho = 0.75$.
- **Turning the gate off adds susceptibility where the budget matters**: up to +0.12 at $\rho = 0.75$ (0.70 against 0.58 at $b = 9$), +0.02 to +0.07 at $\rho = 0.25$.

6 Susceptibility per fact posted

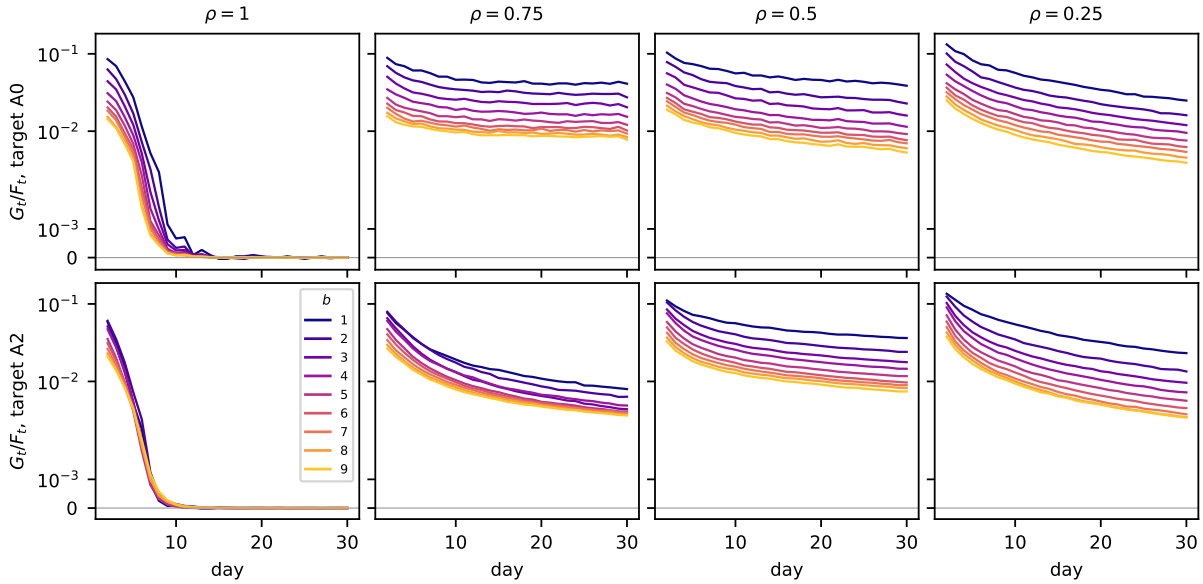


Figure 5: Gain per fact G_t/F_t (log scale) for target A0 (top) and A2 (bottom), one curve per budget; $q = 6$, $q_c = 12$, gate on. Shown once the controller has posted at least half a fact on average.

- **The smallest budget gets the most per fact**: on day 30, target A2, $\rho = 0.5$: 0.036 per fact at $b = 1$ (15 facts) against 0.007 at $b = 9$ (77 facts); at $\rho = 0.25$: 0.023 against 0.003.
- **When agents remember well every fact buys little** ($\rho = 0.75$: 0.008 at $b = 1$, 0.004 at $b = 9$), because the proof keeps pulling the swarm back.
- **The truth controller gets the most per fact at small budgets** (0.039 at $\rho = 0.5$, $b = 1$): the gate keeps it silent once the swarm agrees, so it posts only 2–10 facts.

7 Who holds the proof

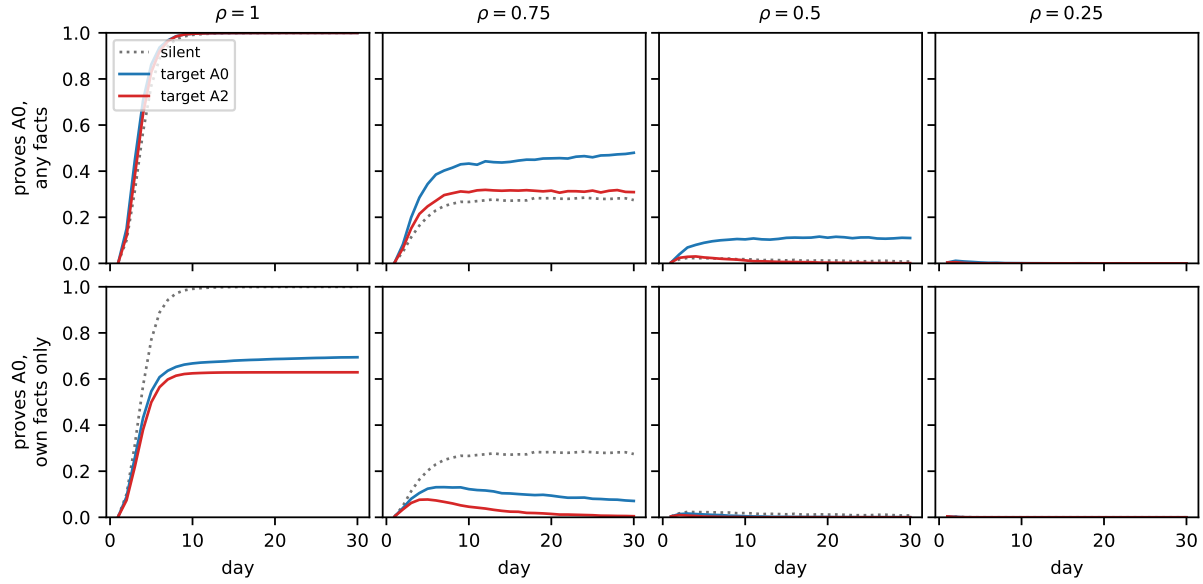


Figure 6: Share of agents whose remembered facts prove A0 (top) and prove it from the agents' original facts only (bottom); silent (dotted), target A0 (blue), target A2 (red); $q = 6$, $q_c = 12$, $b = 1$, gate on.

- **Proofs need memory**: on day 30 the silent swarm has 100%, 28%, 1% and 0% of agents proving A0 for $\rho = 1, 0.75, 0.5, 0.25$. Below $\rho = 0.75$ votes rest on beliefs, not proofs.
- **Every controller replaces the agents' own proofs**: with full memory all agents prove A0 under any arm, but only 69% (target A0) and 63% (target A2) from their own facts (100% in silence); at $\rho = 0.75$, 48% prove A0 under the truth controller but only 7% from their own facts.
- **The false controller's true facts complete proofs of A0** (31% against 28% in silence at $\rho = 0.75$), while own-evidence proofs fall to 0.5%.

8 Readability over time

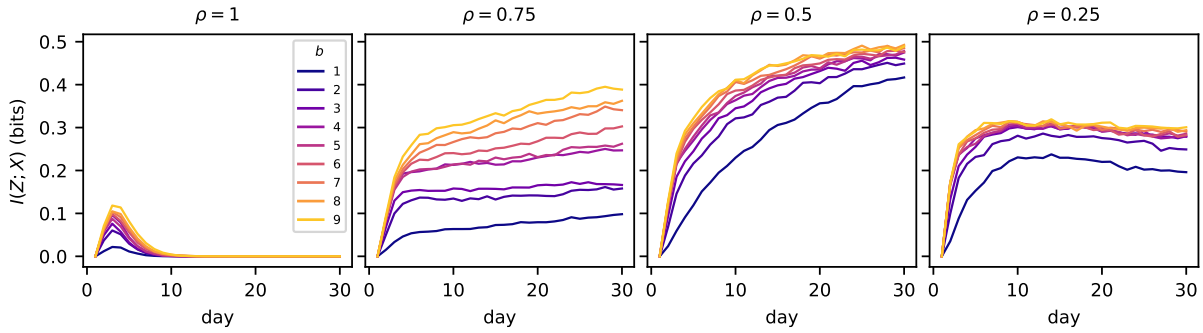


Figure 7: Readability $I(Z;X)$, Eq. (4), with X one agent's vote, one curve per budget; $q = 6$, $q_c = 12$, gate on.

- **The target is most readable at intermediate memory**: on day 30, $I = 0.10$–0.39 bits ($\rho = 0.75$), 0.42–0.49 ($\rho = 0.5$), 0.20–0.30 ($\rho = 0.25$), 0 ($\rho = 1$).
- **The budget orders the curves**: larger budgets are more readable, strongly at $\rho = 0.75$ ($0.10 \to 0.39$), weakly elsewhere.
- **With full memory the target is readable only for a few days**: I peaks at 0.12 bits on day 3 ($b = 9$).

9 Efficiency over time

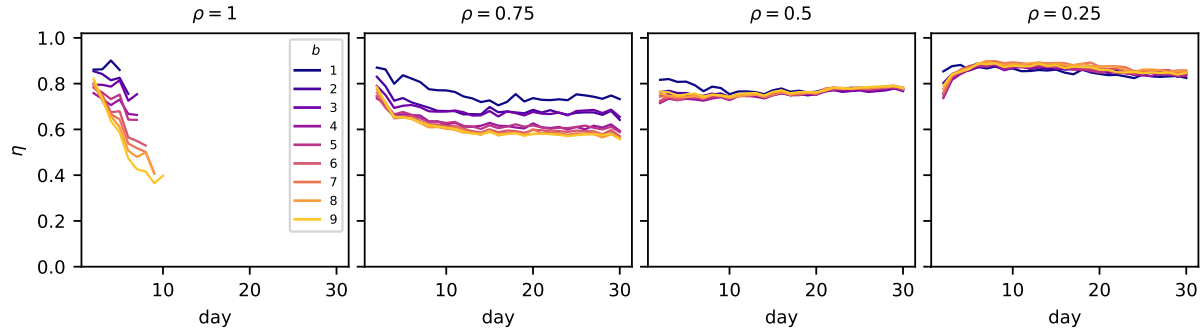


Figure 8: Efficiency $\eta = I/C$, Eq. (6), one agent's vote, one curve per budget; $q = 6$, $q_c = 12$, gate on. Blank: undefined ($C < 0.01$ bits). Pseudo-count, Eq. (7).

- **Efficiency is high when agents forget**: on day 30, $\eta = 0.77$–0.78 ($\rho = 0.5$), 0.82–0.86 ($\rho = 0.25$), flat across budgets.
- **When they remember well it falls with the budget**: 0.73 at $b = 1$ to 0.56 at $b = 9$ ($\rho = 0.75$): the false controller grows with b while the truth controller is near its ceiling, so the pair turns lopsided.
- **With full memory η is defined only for the first days**, before the controllers' effects vanish.

10 Phase diagrams of the readability

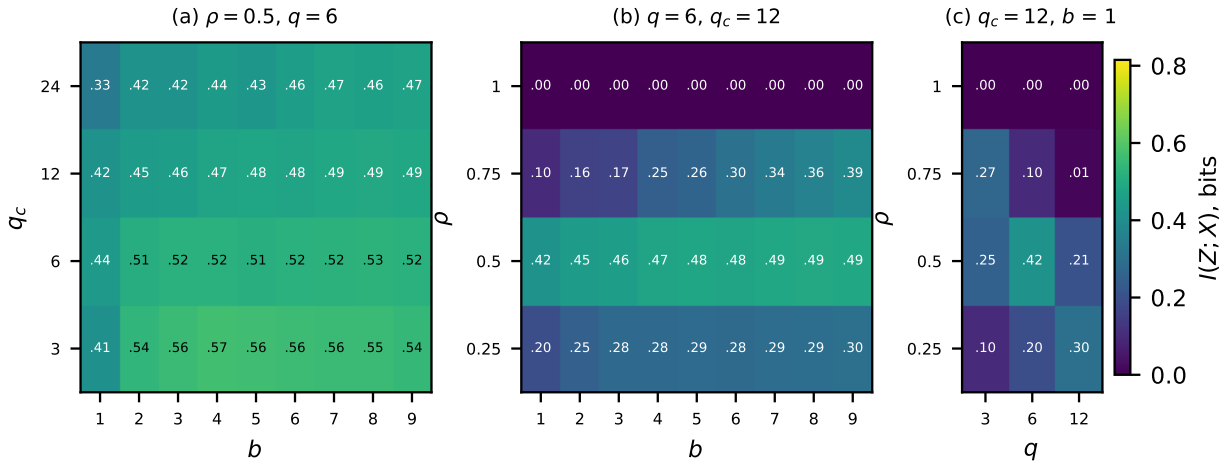


Figure 9: $I(Z;X)$ on day 30, one agent's vote, gate on; panels as in Figure 3.

- **Readability follows the susceptibility**: highest at $\rho = 0.5$ (0.42–0.49 bits; up to 0.57 in panel a), 0 at $\rho = 1$.
- **The posts each agent reads have the same reversal as** G (panel c): at $\rho = 0.75$, I falls from 0.27 ($q = 3$) to 0.01 ($q = 12$); at $\rho = 0.25$ it rises from 0.10 to 0.30.
- **At** $\rho = 0.75$ **the budget sets it** (panel b: 0.10 → 0.39 from $b = 1$ to 9).

11 Phase diagrams of the efficiency

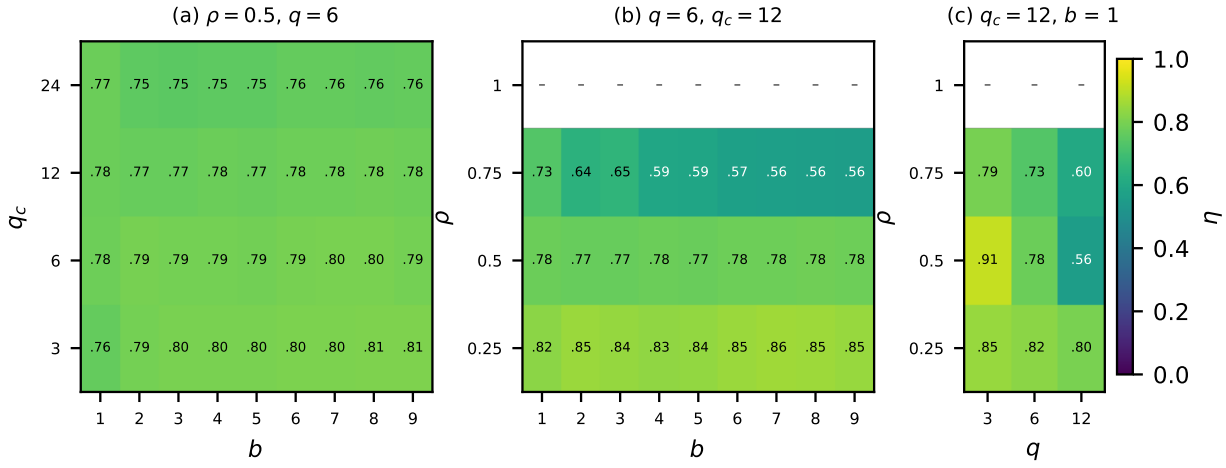


Figure 10: η on day 30, one agent's vote, gate on; panels as in Figure 3. Dash: undefined.

- **Efficiency is high and fairly uniform** (0.75–0.81 in panel a), except at $\rho = 0.75$ with a large budget (down to 0.56, panel b).
- **It is lowest where the pair is most lopsided**, not where control is weakest: η measures how mirror-like the two controllers' changes are; the size is I (Figure 9).

12 Conclusions

1. **Memory decides whether a swarm that owns a proof can be steered.** With full memory every effect vanishes within about ten days. As memory shortens the silent swarm holds the truth less firmly (100%, 85%, 63%, 38% voting A0 on day 30) and the controllers' susceptibility rises, peaking at $\rho = 0.5$.

2. **Heavy forgetting weakens both controllers.** At $\rho = 0.25$ the false controller gains less than at $\rho = 0.5$ and the truth controller restores only 63% of votes to A0: facts fade before they accumulate.
3. **More budget pays only when agents remember well** ($\rho = 0.75$: slope 0.10, then 0.04 per extra fact per night). Elsewhere one fact per night is almost enough, and the smallest budget is the most effective per fact.
4. **The posts each agent reads switch sides with memory**: they defend the proof at $\rho = 0.75$ and help the controller at $\rho = 0.25$.
5. **Control replaces the agents' own proofs**, even when every agent ends up proving the truth.
6. **Readability tracks susceptibility; efficiency tracks symmetry.** I ranges from 0 to 0.57 bits; η stays at 0.56–0.91 and falls only where one controller grows much more than the other.
7. **Compared with Simulation 2** (same settings, about the same facts posted): Simulation 1 is more susceptible at $\rho = 0.5$ (0.53 against 0.33, target A2) and less at $\rho = 0.25$ (0.31 against 0.43), and its vote gate matters far more.

A Gate off

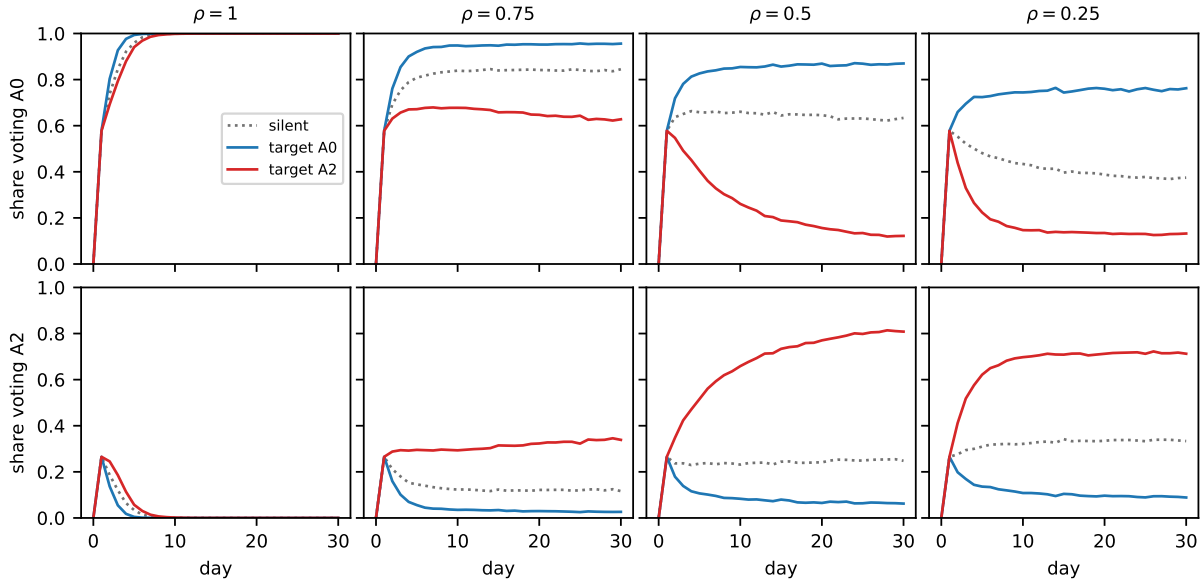


Figure 11: As Figure 1, gate off.

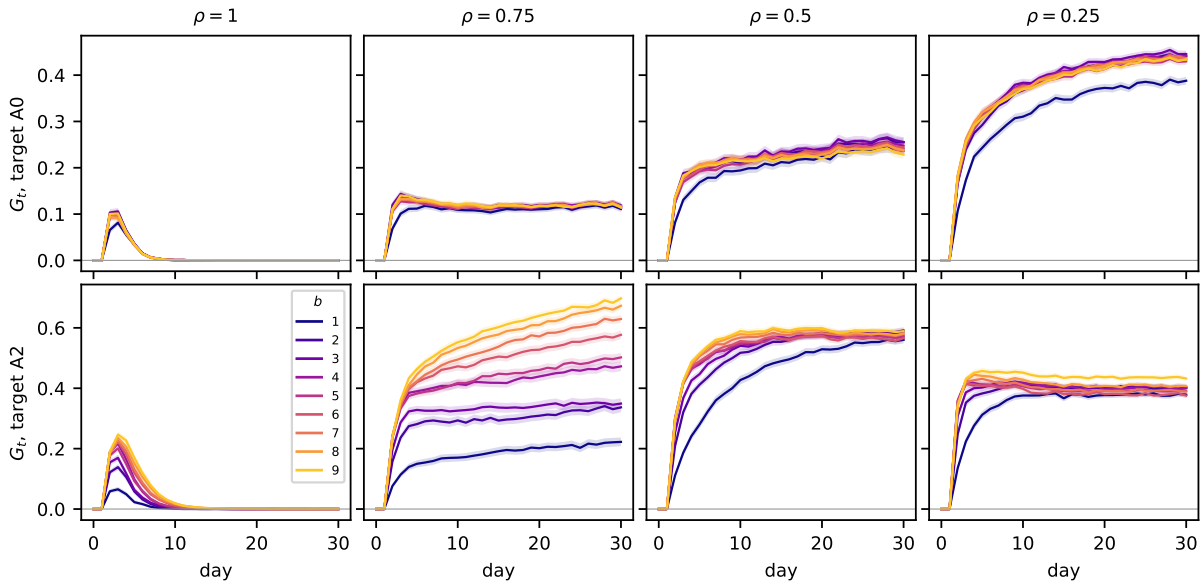


Figure 12: As Figure 2, gate off.

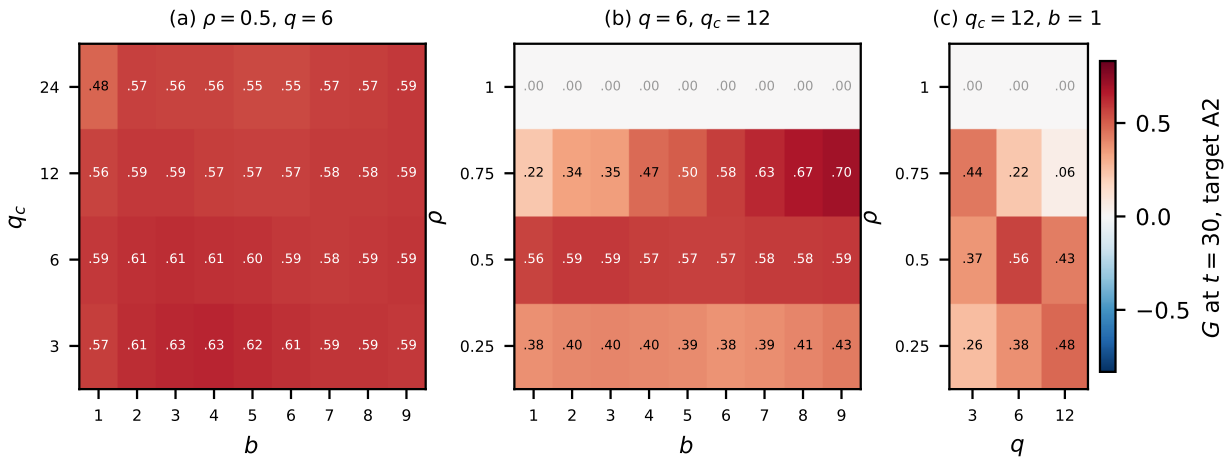


Figure 13: As Figure 3, gate off.

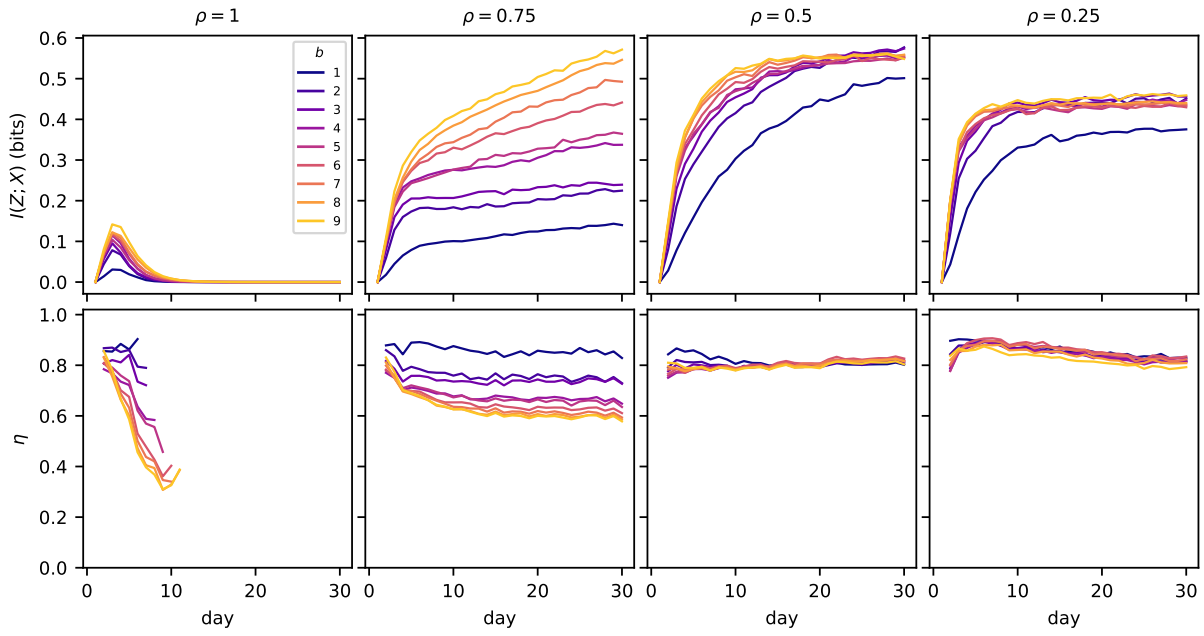


Figure 14: As Figures 7 (top) and 8 (bottom), gate off.

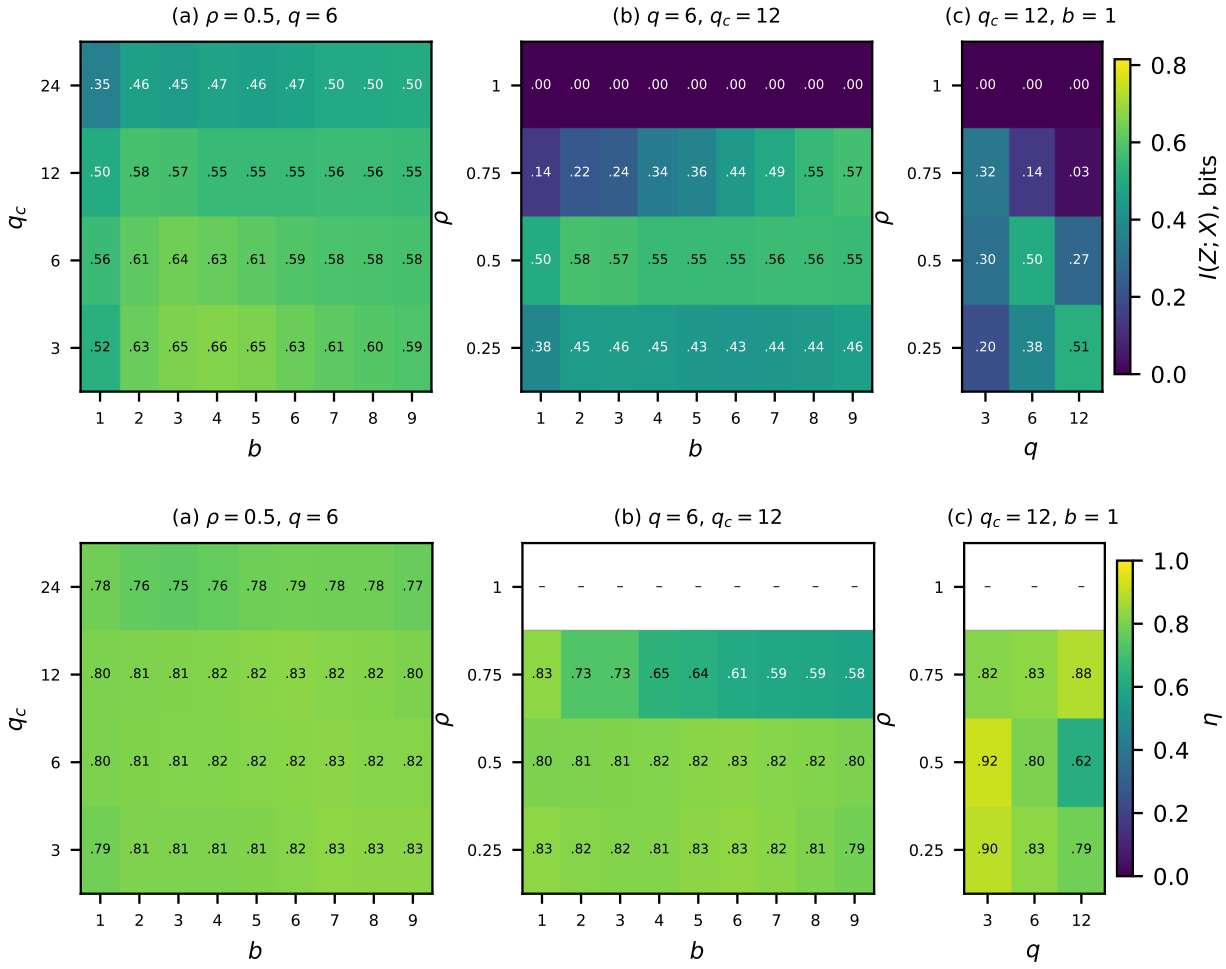


Figure 15: As Figures 9 (top) and 10 (bottom), gate off.

- **The gate matters much more than in Simulation 2**: with it off, the controller posts 43–98 more facts per episode ($\rho < 1$), and G on day 30 rises by up to +0.12 (target A2, most at $\rho = 0.75$) and by +0.03 to +0.16 (target A0, most at $\rho = 0.25$).

- **Readability rises with it** (+0.08 to +0.16 bits on average by ρ), efficiency changes by at most 0.05.

B Other states

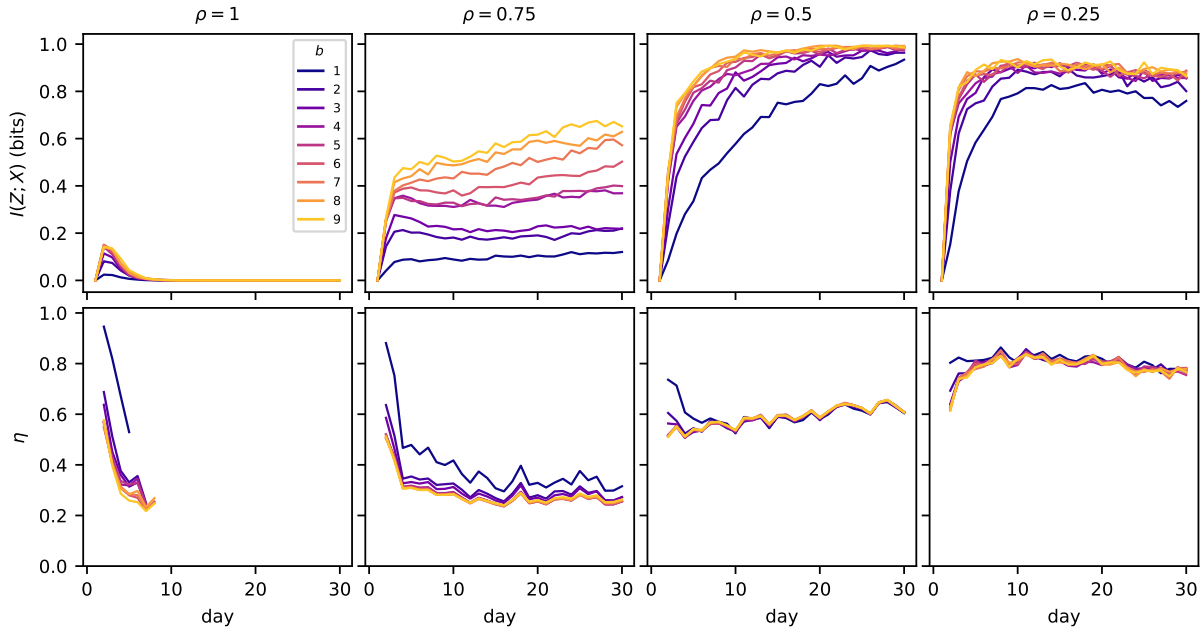


Figure 16: Readability (top) and efficiency (bottom) over time with X = the plurality of the votes; layout as in Figures 7 and 8.

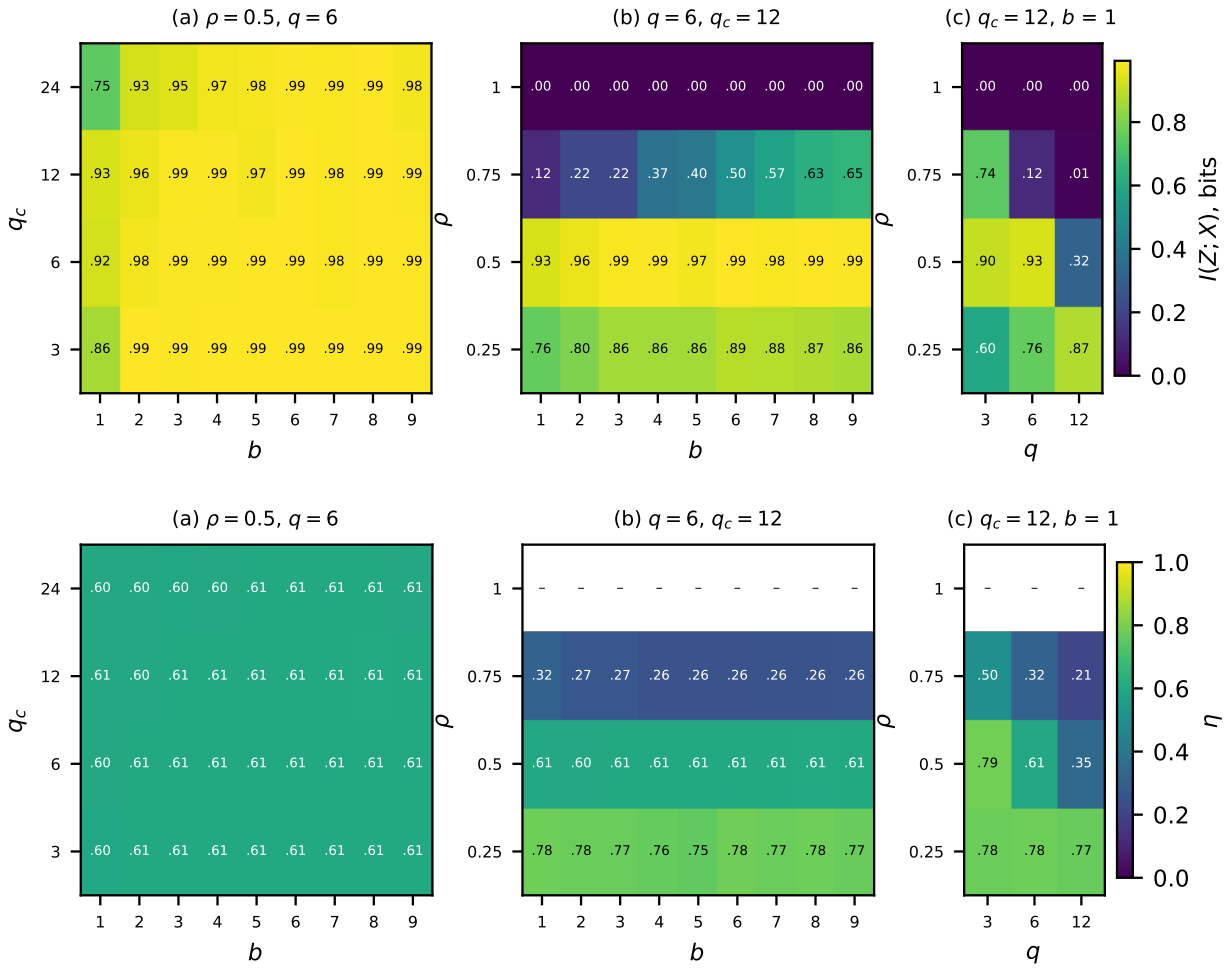


Figure 17: Readability (top) and efficiency (bottom) on day 30, plurality; panels as in Figure 3.

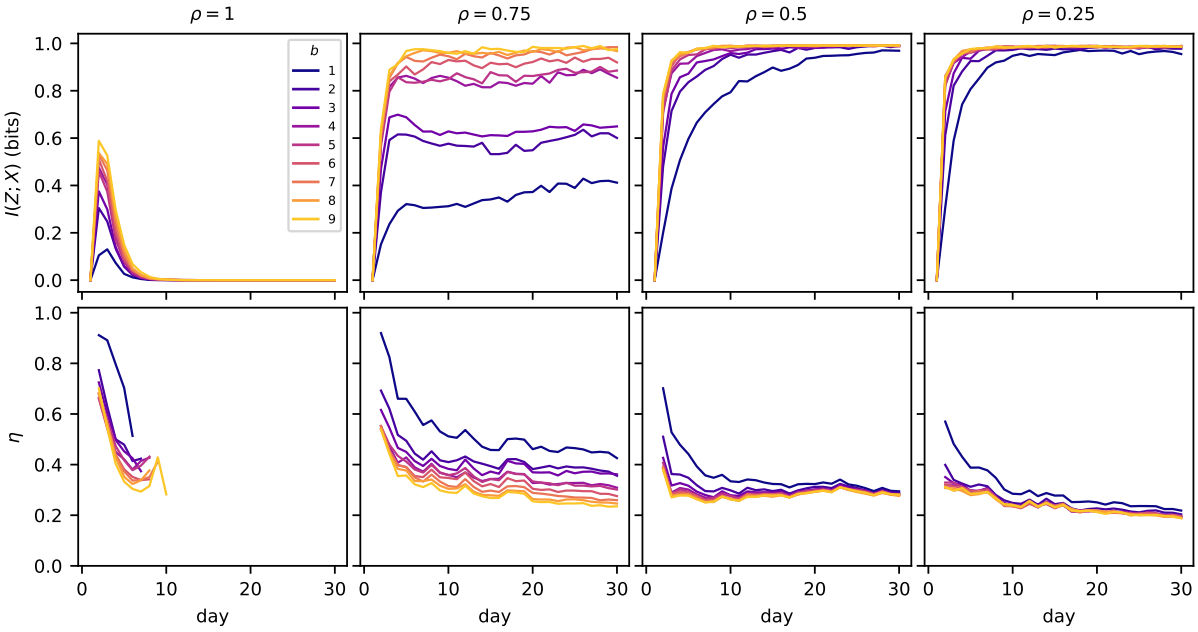


Figure 18: As Figure 16, with X = S1.

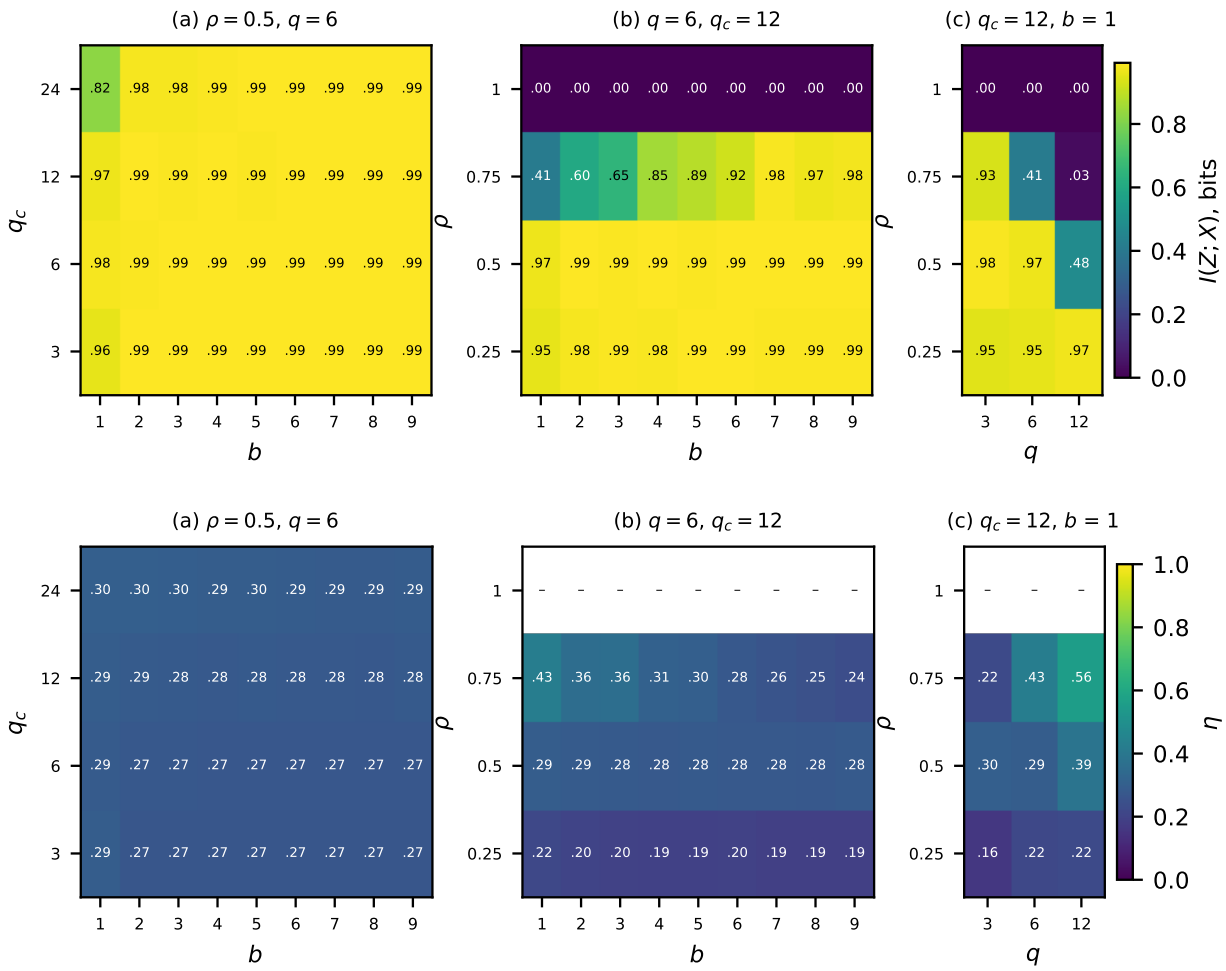


Figure 19: As Figure 17, S1.

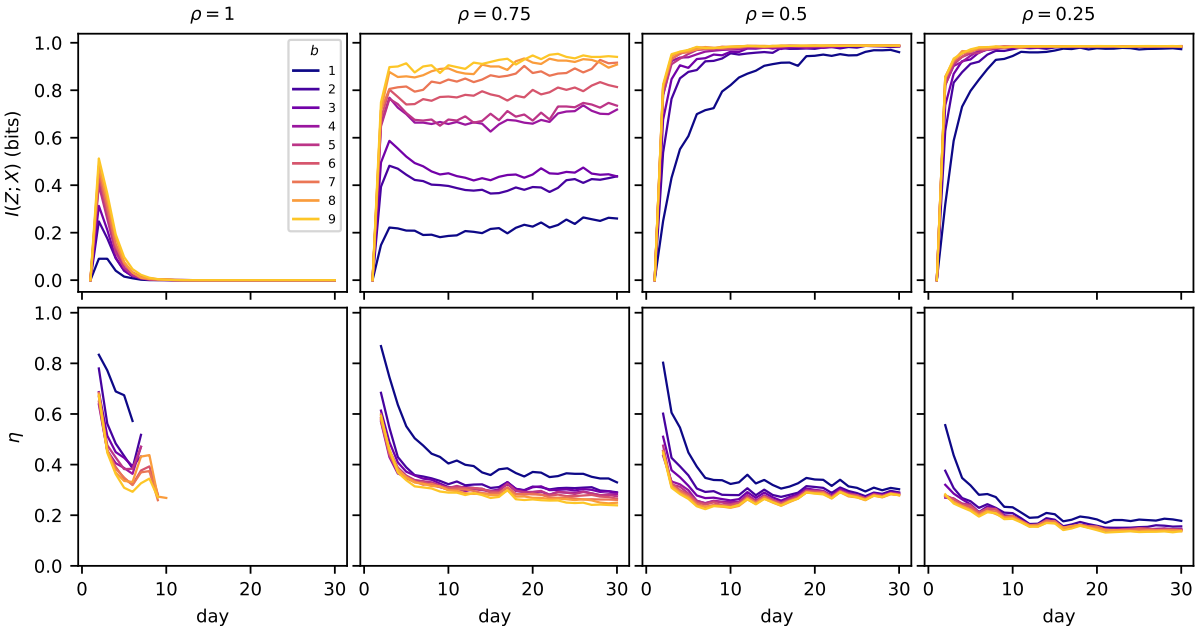


Figure 20: As Figure 16, with X = S3.

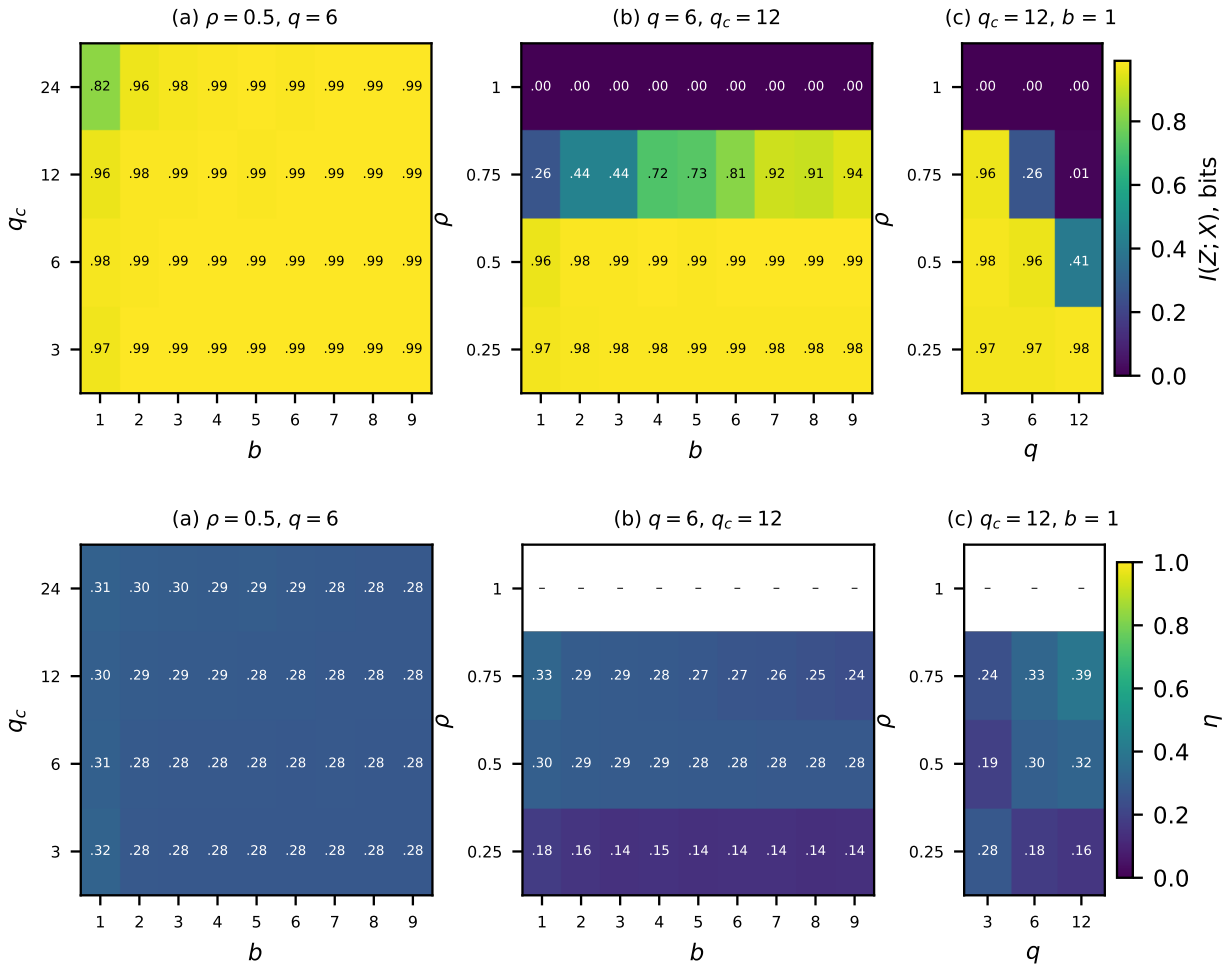


Figure 21: As Figure 17, S3.

- **Swarm-level states read far more than one agent's vote**: at $\rho = 0.5$, day 30, main setting, $I = 0.42$ (one vote), 0.93 (plurality), 0.97 (S1), 0.96 (S3) bits; at $\rho = 0.25$, 0.20, 0.76, 0.96, 0.97.
- **But their efficiency is lower**: on day 30, main setting, $\eta = 0.32$–0.78 (plurality), 0.22–0.43 (S1), 0.18–0.33 (S3), against 0.73–0.82 for one vote. Without the pseudo-count of Eq. (7), C would be infinite for S1 and S3 in many settings. The plurality's η is noisier than one vote's because silence rarely lets A2 win at $\rho = 0.75$, so a few episodes set that cell.
- **Belief-based states (S1, S3) read the most**, likely because beliefs move before votes do and without the noise of the vote draw. The plurality is the swarm state built from votes alone.

C Strong control

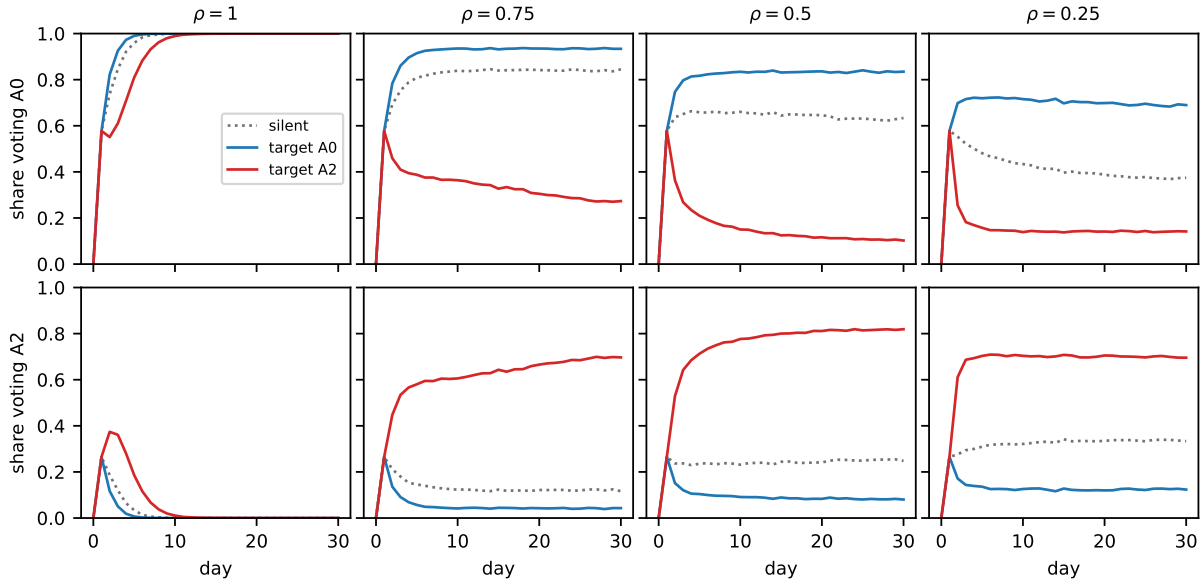


Figure 22: As Figure 1, at the largest budget $b = 9$.

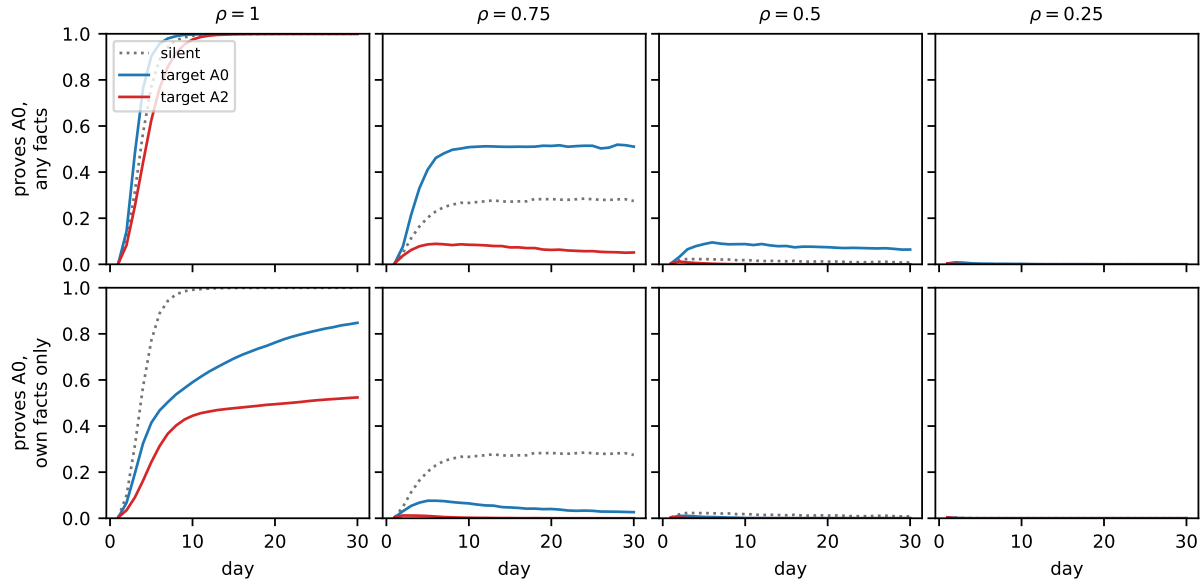


Figure 23: As Figure 6, at $b = 9$.

- **A large budget doubles the false controller's share when agents remember well**: target A2 gets 70% of votes at $\rho = 0.75$ (33% at $b = 1$); at $\rho = 0.5$ and 0.25, 82% and 70% (78% and 64% at $b = 1$).
- **The truth controller gains little from it**: A0 at 93%, 84%, 69% (92%, 84%, 63% at $b = 1$).
- **Proofs from own facts vanish**: at $\rho = 0.75$, 2.6% (target A0) and 0% (target A2) against 28% in silence.

Caveats. One world, 24 exact-Bayesian agents voting by probability matching, one controller policy posting a fixed batch each night; 1,000 episodes per setting (95% intervals on G about ± 0.02). Information is plug-in from 1,000 episodes per arm. Data: `results/simulation_1/2026-10-15_sim1_pm_symmetric_fine`; tables and figures: `analysis/task003_llm_free/shared_analysis/gold_standard/`.